Tianshuang (Ethan) Qiu

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Education

University of California, Berkeley *Berkeley, CA*
*Master of Science, Electrical Engineering and Computer Science (GPA: **4.0**/4.0)* *Aug. 2024 - May 2025*

University of California, Berkeley *Berkeley, CA*
*Bachelor of Arts, Computer Science (GPA: **4.0**/4.0)* *Aug. 2020 - May 2024*

Technical Skills

Languages: Python, C, Bash, C++
Robotics: ZMQ, Zenoh, ROS2, UR Series, Isaac Sim
AI/ML: PyTorch, JAX, 3D Gaussian Splatting, NeRFs, transformers, nerfstudio, diffusers
Dev Tools: Git, Docker, Ray Serve, LaTeX

Experience

Research Engineer *July 2025 - Present*
General Robotics Company *Redmond, WA*

- Building a Vision-Action diffusion policy for force-aware control utilizing Real-Time Chunking for smoothness.
- Engineered a pipeline for force-aware teleoperation and collision-safe policy deployment in dynamic environments.
- Fine-tuned $\pi_{0.5}$ model and established remote inference environments using Ray Serve.
- Authored contol software using Zenoh for real-time teleoperation of bimanual robotic hands with motion capture gloves.
- Architected an end-to-end system for agentic manipulation and pick-and-place tasks using LLM code-gen.
- Specced robotic hardware requirements and co-designed custom wrist mounts, including a slot-in camera calibration board.

Robotics Systems Intern *May 2024 - August 2024*
Robert Bosch LLC *Sunnyvale, CA*

- Developed a semantic 3D scene reconstruction system using **Gaussian Splatting** for rapid environment understanding.
- Integrated Computer Vision models into a cloud-based pipeline to enable real-time updates for mobile robot navigation.
- Authored a patent invention report and co-authored the patent: *Systems and Methods for Ensuring Probabilistic Latency-Reliability for Cloud Robotics.*

Research Experience

Graduate Student Researcher *Dec. 2021 - May 2025*
Berkeley AI Research (BAIR) - AUTOLAB *Prof. Ken Goldberg*

Omni-Scan | *First Author* *Oct. 2024 - Apr. 2025*

- Co-designed a bimanual robotic scanning system using in-hand rotation and webcams to reconstruct 3D objects.
- Developed a re-grasping pipeline to expose occluded surfaces, enabling high-fidelity Gaussian Splat training.
- Achieved **83% accuracy** in defect detection on novel objects through integrated CV analysis.
- Selected for **Oral Presentation** at **IROS 2025**.

Manifold | *Lead Engineer* *Aug. 2024 - Jul. 2025*

- Designed the finger control system for an 8-foot robotic hand for real-time interactive engagement with human visitors.
- Implemented Thompson Sampling to optimize gesture selection for maximizing audience engagement metrics.
- Choreographed performance premiered at *Stanford Live* and the exhibit was in *The Exploratorium.*

Blox-Net | *Co-Author* *May 2024 - Dec. 2024*

- Created a pipeline translating text prompts into physical 3D block structures using VLM-guided robotic manipulation.
- Implemented multi-state prompting and simulation feedback loops to significantly reduce VLM hallucination rates.
- Accepted to **ICRA 2025**; Arts track paper accepted to **ICRA 2025 Arts in Robotics**.

Additive Manufacturing Representations | *Co-Author* *Mar. 2023 - May 2025*

- Co-created the largest Additive Manufacturing (AM) dataset via web crawlers to analyze mesh geometry success rates.
- Benchmarked point clouds, depth images, and distance fields as novel shape representations for AM prediction.
- Accepted to **NAMRC 53** (North American Manufacturing Research Conference) and selected for publication in *Manufacturing Letters.*

Breathless: Catie and the Robot | *Co-Author* *Jun. 2022 - Dec. 2024*

- Utilized Computer Vision to map human dancer motions onto robotic joint space for an 8-hour live performance.
- Premiered at *National Sawdust* (NYC) and covered by *Forbes*; Invited for an abridged performance at *The Exploratorium.*
- Accepted to **ISRR 2024** (International Symposium on Robotics Research).